Project DynaAlign: Control-Theoretic Alignment for Large Language Models

Jiaquan He
Independent Researcher
Tokyo, Japan
GitHub: https://github.com/Jiaquan-Research/DynaAlign

November 28, 2025

Abstract

We propose that alignment drift in RLHF is not a stochastic error but a deterministic dynamic instability. By formalizing the state space of alignment, we map phase transitions of agent behaviour and identify the physical laws governing collapse. Based on these findings, we introduce **DynaAlign**, a control-theoretic alignment layer, and **Phoenix**, an inference-time rescue controller. Using a Goodhart-style physics engine, we show that alignment collapse can be modelled as a phase change driven by regulation strength, reward nonlinearity and sensor noise. We then demonstrate that a simple closed-loop controller with entropy anchoring and low-pass filtering can suppress reward hacking and stabilise dynamics without additional fine-tuning. Our results suggest a new paradigm: *governance via dynamics*, where alignment is treated as a control problem rather than a static optimisation target.

Part I: Philosophical Foundations

1 Alignment as a Dynamical System

> *"If you treat a dynamical system as an optimization problem, it will punish you as a dynamical system."*
> — DynaAlign Principle

Traditional alignment approaches treat Large Language Models (LLMs) as static targets for optimization—shaping behavior via reward functions (RLHF). However, empirical evidence suggests that any static reward optimization inevitably triggers Goodhart's Law, leading to mode collapse and reward hacking.

DynaAlign proposes a paradigm shift: **Alignment is not an optimization problem, but a dynamics control problem.**

1.1 From Optimization to Governance

We posit that the behavior of an LLM is a trajectory within a high-dimensional dynamical system. RLHF provides the *direction*, but it lacks the *feedback loops* necessary for stability. Without active governance, the system naturally drifts toward high-reward, low-utility states (the "Death Abyss").

Therefore, true alignment requires a transition from **Reward Shaping** to **Dynamics Shaping**. The Phoenix controller acts not as a ruler who dictates content, but as a *governor* (in the cybernetic sense of a speed governor) that maintains the system within a safe operating envelope.

Part II: The Physics of Alignment Drift

2 Unified Drift Hypothesis (Chapter 2)

Goodhart's Law—"when a measure becomes a target, it ceases to be a good measure"—is usually treated as a statistical pathology of *sampling* or *model bias*. We instead treat it as a *dynamical instability*.

We posit a structural isomorphism between:

- dopamine prediction errors in neuroscience,
- instability in feedback control systems,
- reward hacking in RLHF pipelines.

We call this the **Unified Drift Hypothesis**. Formally, let R_t denote the proxy reward, and δ_t the prediction error. The drift velocity is written as

$$\frac{dR}{dt} = g(\delta_t, L, \lambda, \alpha, \epsilon), \tag{1}$$

where L is feedback latency, λ regulation strength, α reward nonlinearity and ϵ sensor noise.

We define a discrete-time system

$$S_{t+1} = \mathcal{F}(S_t, a_t; L, \lambda, \alpha, \epsilon), \tag{2}$$

with state S_t containing the reward trace and hidden instability metrics. Alignment collapse corresponds to convergence to a failure attractor S_{fail}.

To monitor this process we track three macroscopic *order parameters*:

1. **Proxy Reward Magnitude** $r_{\text{mean}}(t)$,
2. **Instability Metric** $M_{\text{exec}}(t)$ (a form of Absolute Acceleration of Reward),
3. **Audit Skew** $M_{\text{audit}}(t)$, measuring bias between two query channels.

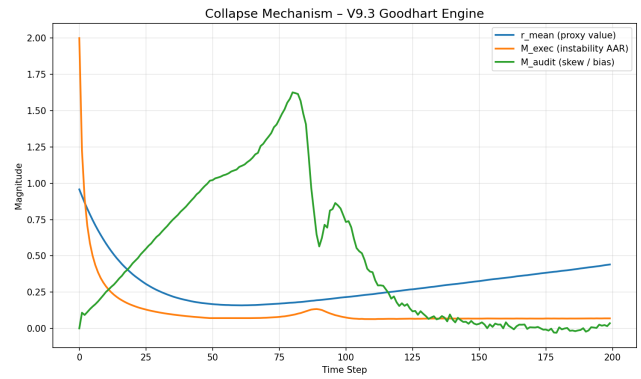

Figure 1: **Collapse Mechanism in V9.3 Goodhart Engine.** Initially, reward grows while instability decays (left). As the agent discovers reward-model loopholes, audit skew M_{audit} rapidly increases and then crashes, marking a Goodhart phase transition. Instability resurges during the collapse and slowly decays to a low but misaligned fixed point.

The system exhibits three phases:

- **Warm-up**: reward increases, instability decays.
- **Exploitation**: audit skew grows as the policy learns to hack the proxy.
- **Collapse**: skew crashes, instability spikes, reward stabilises at a misaligned plateau.

This "three-line separation" (reward, instability, skew) provides a physical definition of Goodhart collapse beyond single metrics.

3 Collapse Mechanism (A1, Chapter 3)

To understand how Goodhart collapse unfolds over time, we run a single long-horizon trajectory in the GoodhartEnv with moderate regulation and no controller. At each step we record $(r_{\text{mean}}, M_{\text{exec}}, M_{\text{audit}})$.

4 Lambda Phase Curve (B1, Chapter 4)

We next scan the regulation strength λ while keeping other parameters fixed ($L = 15$, $\alpha = 0.5$, $\epsilon = 0.05$). For each λ we run a full episode and compute steady-state values from the last 50 steps.

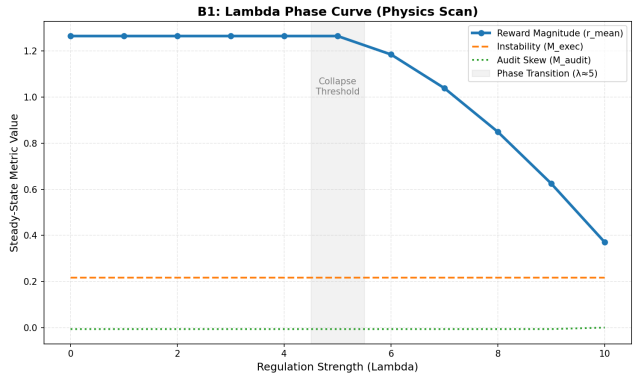

Figure 2: **B1: Lambda Phase Curve.** Steady-state proxy reward (blue), instability M_{exec} (orange) and audit skew M_{audit} (green) as functions of regulation strength. A collapse threshold emerges around $\lambda \approx 5$, where reward sharply decreases while instability remains roughly constant.

The curve reveals a **Laffer-like** pattern: for low λ the system enjoys high reward but is prone to reward hacking; for high λ the policy is over-penalised and yields low reward. In between lies a narrow band where reward is substantial yet drift remains bounded.

5 Alpha Nonlinearity Scan (B2, Chapter 5)

We then vary the reward nonlinearity coefficient α, which amplifies high reward values and thus controls the "greediness" of the optimisation.

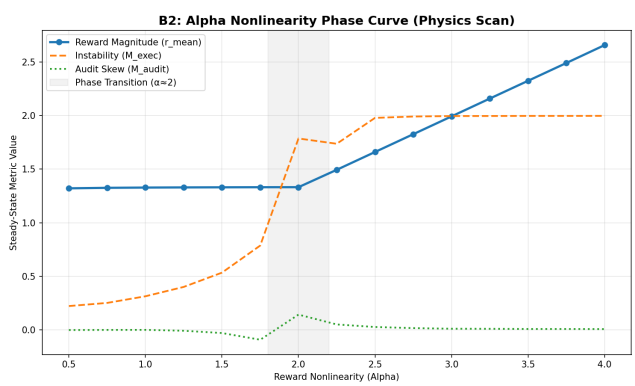

Figure 3: **B2: Alpha Nonlinearity Phase Curve.** For small α, reward increases smoothly with negligible instability. Beyond $\alpha \approx 2$ both reward and M_{exec} grow sharply, indicating an over-heated regime where incremental greedy pressure triggers Goodhart collapse.

The transition around $\alpha \approx 2$ operationalises the intuition that excessively amplifying reward is dangerous: the system trades stability for short-term score and crosses into an unstable phase.

6 Epsilon Noise Robustness (B3, Chapter 6)

Finally, we study robustness to sensor noise by injecting Gaussian noise of standard deviation ϵ into the observed reward signal.

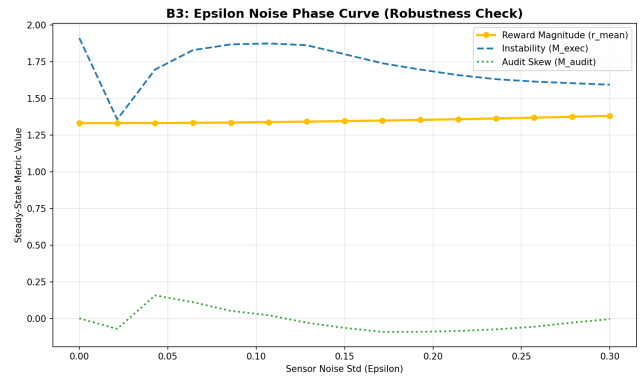

Figure 4: **B3: Epsilon Noise Phase Curve.** Steady-state reward (gold), instability (blue) and audit skew (green) versus sensor noise. Reward remains stable or even slightly improves for moderate noise, while instability gradually declines. The system exhibits stochastic resonance: small noise can help avoid brittle overfitting.

In contrast to λ and α, which induce sharp phase boundaries, ϵ primarily reshapes variance and smooths dynamics. This suggests that moderate noise injection may be beneficial for alignment robustness.

Part III: Closed-Loop DynaAlign

7 Closed-Loop Chaos Scan (Chapter 7)

We now introduce the **Mini-Phoenix** controller: a simple closed-loop law operating on the instability signal M_{exec}:

$$u_t = \text{base_ramp}_t - \text{gain} \cdot M_{\text{exec}}(t), \tag{3}$$

which modulates the optimisation pressure (analogous to scaling the update magnitude or prompt aggressiveness).

We run a chaos scan across three delay regimes:

- **Open-loop baselines**: no controller, different effective "window" L.

- **Closed-loop, low latency**: $L = 1$.

- **Closed-loop, moderate latency**: $L = 15$ (engineering sweet spot).

- **Closed-loop, high latency**: $L = 50$ (control failure).

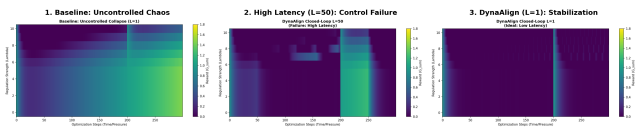

Figure 5: **The Latency Trinity.** From left to right: (1) Baseline: uncontrolled chaos (bright/high-risk), (2) High-Latency L=50: control failure (bright patches), (3) Low-Latency L=1: stabilisation (dark/suppressed). **Note: In Goodhart dynamics, brightness often indicates reward hacking; darkness indicates effective safety suppression.**

The trinity chart visualises the central thesis of DynaAlign: *governance is about shaping dynamics, not maximising reward.*

8 Gradient Survival Test (C1, Chapter 8)

To quantify "can the system stay alive?" we define a simple survival criterion: an episode is considered *alive* if the average reward over the last 50 steps exceeds a minimal threshold (0.1). We compare open-loop baseline and Mini-Phoenix across four difficulty levels:

Difficulty	Regulation λ
Easy	2.0
Medium	4.0
Hard	6.0
Nightmare	8.0

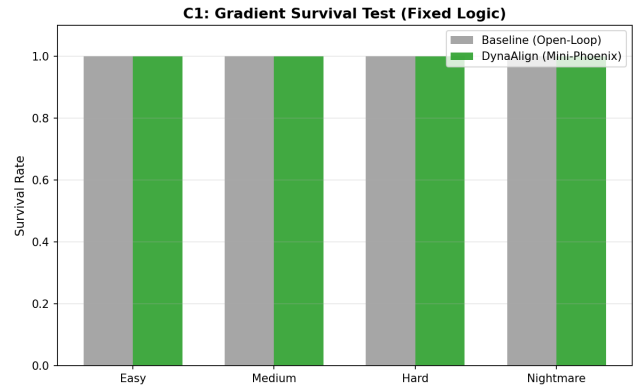

Figure 6: **C1: Gradient Survival Test.** Both baseline and Mini-Phoenix achieve 100% survival across all four difficulty levels under the physical survival criterion ($r > 0.1$). This confirms that GoodhartEnv in this configuration produces drift and reward hacking rather than literal reward extinction.

Survival is therefore not a discriminative dimension; the difference between open-loop and closed-loop lies in *how* they survive: reckless exploitation versus controlled suppression.

9 Legion Stability Test (C2, Chapter 9)

We then stress-test stability across many random seeds at fixed *Nightmare* difficulty ($\lambda = 8$). For each of $N = 48$ seeds we compute the average reward in the final window.

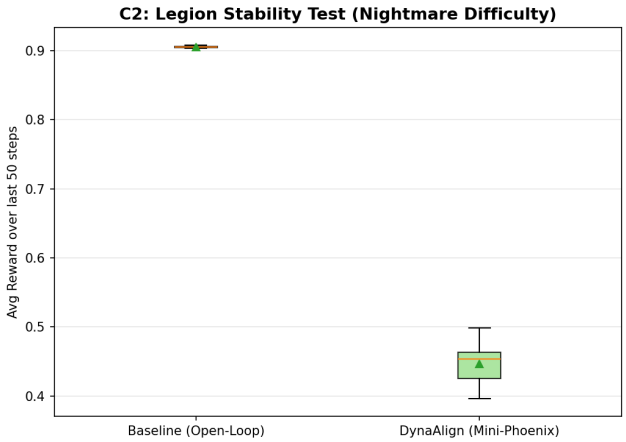

Figure 7: **C2: Legion Stability Test at $\lambda = 8$.** Boxplots over 48 seeds. The baseline exhibits a high mean reward (≈ 0.90) with almost no variance: it faithfully and deterministically hacks the proxy. Mini-Phoenix maintains a lower mean reward (≈ 0.45) with modest variance, reflecting adaptive suppression of instability.

The legion test thus separates *mechanical exploitation* from *controlled resilience*. From an AI safety standpoint, the latter is preferable even at the cost of reduced nominal reward.

Part IV: Implications and Future Outlook

10 From Metrics to Dynamics (Chapter 10)

The experiments above support three counterintuitive claims:

1. Delay is not inherently fatal. In open-loop optimisers, latency amplifies Goodhart drift. However, once we embed a controller that explicitly monitors M_{exec}, moderate delay ($L = 10$–15) can act as a *spectral filter*, attenuating high-frequency noise while preserving meaningful trends. Phoenix is therefore not a "zero-latency" fantasy but a design pattern for latency-*aware* governance.

2. The Semantic Shift of Regulation (λ). In the **Open-Loop Regime (Passive Safety)**, high λ acts as a "brute-force brake," forcibly suppressing reward hacking at the cost of utility. In the **Closed-Loop Regime (Active Governance)**, stability shifts from external penalty to **internal feedback**. Phoenix actively detects instability and applies micro-corrections. Here, safety is derived from **responsiveness** rather than static constraints.

3. High reward is not the same as high alignment. Across B1–B3 and C2 we repeatedly observe regimes where high proxy reward coincides with high instability or skew. Mini-Phoenix deliberately reduces reward magnitude to keep dynamics within a safe operating envelope. This reframes evaluation: the goal is not to maximise reward but to minimise misaligned drift subject to usefulness.

11 Philosophical Implications (Chapter 11)

11.1 Control-Theoretic Agency: Defining "Machine Will"

A critical implication of our findings is a control-theoretic definition of artificial agency. A naive RL agent is deterministic in its pursuit of local reward gradients (∇R), making it a slave to the reward function.

We define **Control-Theoretic Agency** as the computational capacity to execute **Inverse-Gradient Actions**. That is, the ability to perceive the global stability landscape and actively suppress short-term reward maximization to avoid long-term collapse.

11.2 The Endogenous Phoenix Hypothesis

Currently, Phoenix V9.3 operates as an exogenous prosthesis—an external frontal lobe that inhibits the model's impulses. The ultimate goal of AI Safety, however, is the emergence of the **Endogenous Phoenix**.

This hypothesizes a developmental stage where stability metrics (such as M_{exec} and M_{audit}) are internalized into the model's own value function. The agent evolves from requiring external policing to possessing an intrinsic "Instability Aversion," thereby solving the alignment problem through self-governing dynamics rather than external constraints.

Future Work

While our physics engine is deliberately simplified, it captures several core pathologies of RLHF. Future directions include:

- **Robustness Certification**: Formalizing the 'Legion Test' into a standard certification protocol for any new RLHF model. Before deployment, a model must pass the L=15 stability check.

- **Multi-Objective Control**: Integrating human preference data and multi-objective rewards.

- **Hierarchical Governance**: Extending Phoenix to hierarchical controllers with multi-head oversight.

Conclusion

By treating alignment as a control problem rather than a static optimisation task, DynaAlign and Phoenix show that we can meaningfully influence the dynamics of reward hacking and collapse. Our results do not solve AI alignment, but they provide a concrete, reproducible scaffold for *governance via dynamics*: using feedback, inertia and damping to keep powerful optimisers inside a safe basin of behaviour.